Assignment 1: Security ML Pipeline, Model Comparison, and AI-Assisted Analysis

Item	Requirement
Submission file	Assignment1-LastName.ipynb
Datasets	PhiUSIIL Phishing URL Dataset and UNSW-NB15
Required models	Logistic Regression, Decision Tree, Random Forest
AI use	Allowed with required AI Use Statement
Total points	100

Introduction

In Lab 1, you completed a guided phishing URL classification workflow using the PhiUSIIL dataset. In this assignment, you will extend that workflow into a more complete security machine learning analysis using two cybersecurity datasets.

You will inspect both datasets, select meaningful features, create engineered features, train and compare multiple models, tune hyperparameters, evaluate model behavior, and explain the results from a cybersecurity perspective.

Datasets

Dataset	Required Files	Target Column	Class Meaning
PhiUSIIL Phishing URL Dataset	PhiUSIIL_Phishing_URL_Dataset.csv	label	0 = Legitimate URL, 1 = Phishing URL
UNSW-NB15 Intrusion Detection Dataset	UNSW_NB15_training-set.csv UNSW_NB15_testing-set.csv	label	0 = Normal traffic, 1 = Attack traffic

AI Use Policy

You may use AI tools such as ChatGPT, Copilot, Gemini, Claude, or similar tools. However, you are responsible for understanding, testing, correcting, and explaining everything you submit.

Allowed AI use:

- Explaining code errors
- Suggesting debugging strategies
- Explaining Python, Pandas, or scikit-learn functions
- Suggesting feature engineering ideas
- Helping interpret model evaluation results
- Improving grammar and clarity in written explanations

Not allowed:

- Submitting AI-generated work that you do not understand
- Submitting code that you cannot explain
- Fabricating results
- Claiming a model performed better than it actually did
- Copying another student's notebook or report

At the end of your notebook, include an AI Use Statement. Use one of the following formats:

```
AI Use Statement: I used [tool name] to help with [specific task]. I reviewed, tested, and modified the output before including it in my assignment.

AI Use Statement: I did not use AI tools for this assignment.
```

Required Packages

Use the following packages. You may import additional packages if needed.

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split, StratifiedKFold, cross_validate, GridSearchCV
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from sklearn.metrics import classification_report, confusion_matrix, ConfusionMatrixDisplay
from sklearn.metrics import roc_auc_score, RocCurveDisplay
```

Part 1: Dataset Loading and Inspection

Complete this part for both datasets.

1. Load the dataset.
2. Display the first five rows.
3. Display the number of rows and columns.
4. Display column names and data types.
5. Check missing values.
6. Check duplicate rows.
7. Display class counts and class percentages.

Required outputs:

- First five rows for each dataset
- Shape of each dataset
- Missing value summary for each dataset
- Duplicate row count for each dataset
- Class distribution table for each dataset

Part 2: Class Distribution Visualization

Create one class distribution plot for each dataset.

- PhiUSIIL class distribution plot
- UNSW-NB15 class distribution plot

Briefly explain whether each dataset is balanced or imbalanced and why class imbalance matters in security detection.

Part 3: Feature Selection

Choose a reasonable set of features for each dataset. Use approximately 8 to 15 features for PhiUSIIL and 8 to 15 features for UNSW-NB15.

For PhiUSIIL, prioritize URL structure, webpage behavior, and URL characteristics. For UNSW-NB15, prioritize network-flow behavior and traffic characteristics.

Required table:

Dataset	Selected Feature	Reason for Selecting It
PhiUSIIL		
UNSW-NB15		

Part 4: Basic Feature Engineering

Create at least two engineered features for each dataset. These features should be based on existing columns and should have a clear security-related purpose.

Examples of useful engineered features include log-transformed length or byte values, ratios, totals, and rate-like features. Do not create random features that have no security meaning.

Required table:

Dataset	Engineered Feature	Formula or Description	Why It May Help
PhiUSIIL			
UNSW-NB15			

Part 5: Feature Distribution and Correlation Analysis

For each dataset, create visual summaries that help you understand selected features before modeling.

- At least two histograms for PhiUSIIL
- At least two histograms for UNSW-NB15
- At least one raw vs. log-transformed comparison if a feature is highly skewed
- One correlation heatmap for PhiUSIIL using selected numeric features
- One correlation heatmap for UNSW-NB15 using selected numeric features

Briefly explain which features appear skewed, which features appear highly correlated, and whether any features may be redundant.

Part 6: Data Cleaning and Preparation

Complete data preparation for both datasets before training models.

1. Create the feature matrix and target vector.
2. Split the data into training and testing sets.
3. Use stratification when splitting.
4. Handle missing values using training data only.
5. Scale numeric features when needed.
6. Encode categorical features in UNSW-NB15 if categorical columns are used.

Important rule: do not fit scaling, encoding, or imputation on the full dataset before splitting. The test set must not influence training decisions.

Part 7: Baseline Model Training

Train and evaluate the following models on both datasets:

- Logistic Regression
- Decision Tree
- Random Forest

For each model and dataset, generate a classification report, confusion matrix, and metric summary.

Required table:

Dataset	Model	Accuracy	Precision	Recall	F1
PhiUSIIL	Logistic Regression				
PhiUSIIL	Decision Tree				
PhiUSIIL	Random Forest				
UNSW-NB15	Logistic Regression				
UNSW-NB15	Decision Tree				
UNSW-NB15	Random Forest				

Part 8: Stratified k-Fold Cross-Validation

Use stratified 5-fold cross-validation to evaluate each required model on both datasets. Report mean performance across folds.

Required metrics:

- Accuracy
- Precision
- Recall
- F1-score

Required table:

Dataset	Model	CV Accuracy	CV Precision	CV Recall	CV F1
PhiUSIIL	Logistic Regression				
PhiUSIIL	Decision Tree				
PhiUSIIL	Random Forest				
UNSW-NB15	Logistic Regression				
UNSW-NB15	Decision Tree				
UNSW-NB15	Random Forest				

Briefly explain why stratified cross-validation is useful for cybersecurity datasets and whether cross-validation results agree with the test-set results.

Part 9: Hyperparameter Tuning

Tune the Decision Tree and Random Forest models for both datasets. Use F1-score as the tuning metric because both phishing and attack detection require attention to false positives and false negatives.

For Decision Tree, tune parameters such as max_depth, min_samples_split, and min_samples_leaf. For Random Forest, tune parameters such as n_estimators, max_depth, min_samples_leaf, and max_features.

Required outputs:

- Best parameters for each tuned model
- Best cross-validation F1 score for each tuned model
- Test-set classification report after tuning
- Test-set confusion matrix after tuning

Required table:

Dataset	Tuned Model	Best Parameters	Best CV F1	Test F1
PhiUSIIL	Decision Tree			
PhiUSIIL	Random Forest			
UNSW-NB15	Decision Tree			
UNSW-NB15	Random Forest			

Part 10: ROC-AUC Analysis

Compute ROC-AUC for models that can produce probability scores. At minimum, include Logistic Regression and Random Forest for both datasets.

- Report ROC-AUC score.
- Plot ROC curves.
- Briefly explain what ROC-AUC measures.
- Briefly explain why ROC-AUC alone is not enough for imbalanced security datasets.

Required table:

Dataset	Model	ROC-AUC
PhiUSIIL	Logistic Regression	
PhiUSIIL	Random Forest	
UNSW-NB15	Logistic Regression	
UNSW-NB15	Random Forest	

Part 11: Security Interpretation

Write a short interpretation for each dataset. Your explanation should be clear, specific, and connected to the actual results shown in your notebook.

For PhiUSIIL:

- Identify the best model.
- Identify the model with the best phishing recall.
- Explain whether false positives or false negatives are more dangerous in phishing detection.
- State whether you would deploy the best-performing model and explain why.

For UNSW-NB15:

- Identify the best model.
- Identify the model with the best attack recall.
- Explain whether false positives or false negatives are more dangerous in intrusion detection.
- State whether you would deploy the best-performing model and explain why.

Part 12: AI Use Statement

End your notebook with a clear AI Use Statement. State which tool you used, what you used it for, and how you verified or modified the output. If you did not use AI, state that directly.

What to Submit

Submit one Jupyter Notebook file only:

```
Assignment1-LastName.ipynb
```

The notebook must include code, outputs, plots, tables, written explanations, and the AI Use Statement. No separate PDF is required.

Grading Rubric

Component	Points
Dataset loading and inspection for both datasets	10
Class distribution and exploratory visualizations	10
Feature selection and feature engineering	15
Data cleaning and preparation	15
Baseline model training and test-set evaluation	15
Stratified k-fold cross-validation	10
Hyperparameter tuning for Decision Tree and Random Forest	15
ROC-AUC analysis and confusion matrix interpretation	5
Security interpretation and AI Use Statement	5
Total	100

Important Notes

- Do not use the test set during preprocessing decisions, tuning, or cross-validation.
- Do not report only accuracy.
- Precision, recall, F1-score, confusion matrix, and ROC-AUC must all be considered.
- A model with high accuracy but low phishing or attack recall may still be a poor security model.
- Your explanations must be written in your own words.
- You must be able to explain your code if asked.